ALPHA HANDBOOK

Complete R&D guide · Strategy Lab

2018-2026 backtest · 9 coins · 4h timeframe · Hyperliquid maker fees

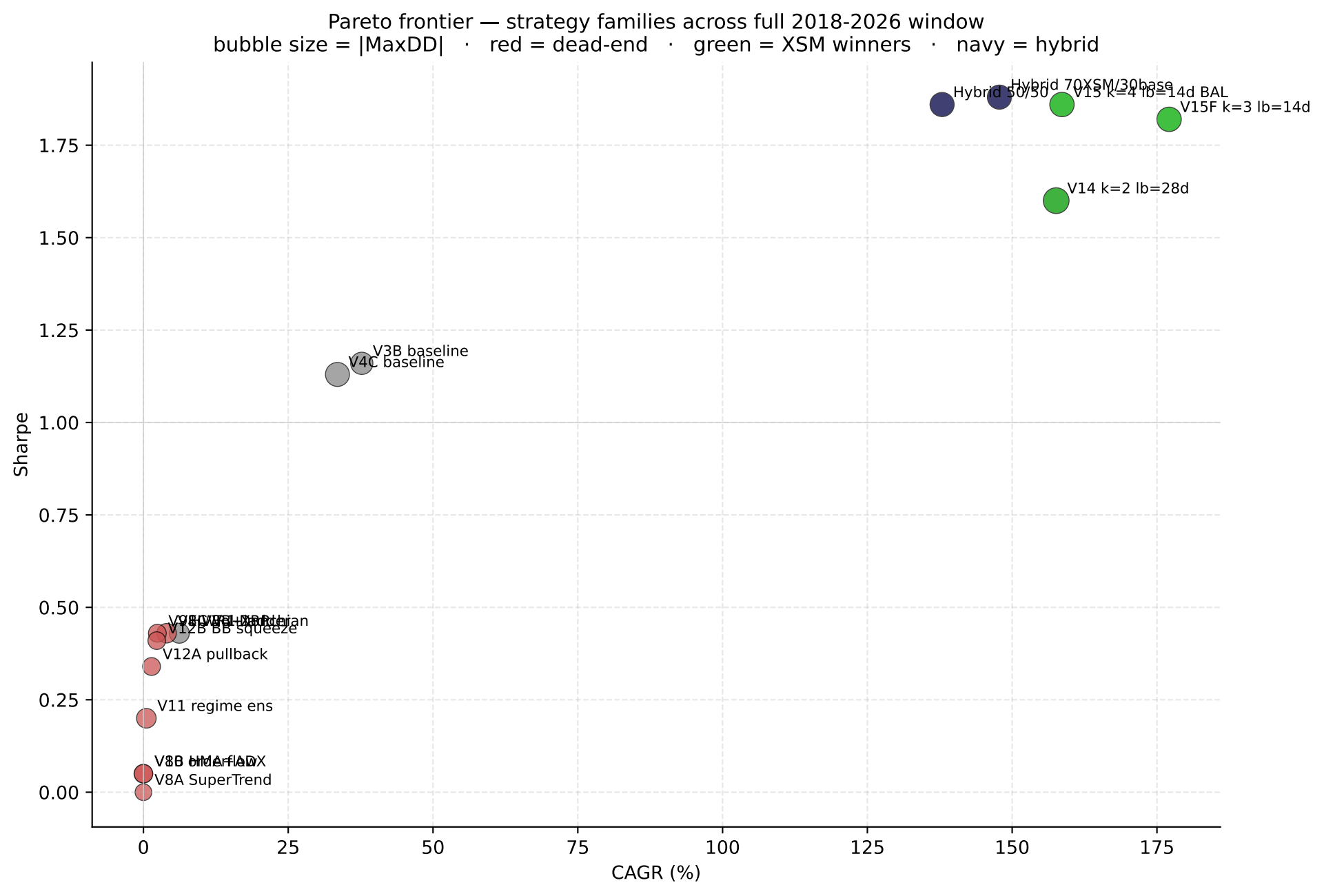
Headline finding: Cross-sectional momentum (XSM) beats time-series trend by ~4x CAGR.

Hybrid portfolio (70% XSM + 30% Trend) hits best-ever Sharpe 1.88 on the 2018-2026 window.

Strategy family	CAGR FULL	Sharpe	MaxDD	Final on $10k
Trend-only baseline	+37.7 %	1.16	-32.9 %	$139,520
XSM k=2 lb=28d (V	+157.6 %	1.60	-58.0 %	$24,393,857
XSM k=4 lb=14d (V	+158.6 %	1.86	-48.3 %	$25,234,358
Hybrid 70% V15 + 3	+147.8 %	1.88	-45.8 %	$17,705,907

Experiments summarised in this handbook:

V7 HWR rule-based high-WR family (BB meanrev / RSI-stoch / pullback)
V8 Novel entries: triple SuperTrend, HMA+ADX, Vol-Donchian
V9 Multi-TP ladder wrappers on V3B/V4C/V2B entries
V10 Orderflow filters (funding / OI / L-S ratio / liquidation cascades)
V11 Regime-switching ensemble (V4C in bull, HWR1 in chop, flat in bear)
V12 Pullback / BB squeeze / higher-lows / NR7 breakout
V14 Cross-sectional momentum — THE BREAKTHROUGH
V15 XSM variant sweep: 70 configs (long-short, composite rank, vol-adj, vol-weighted, grid)
V16 ML-ranked XSM (GradientBoostingRegressor on features)
V17 Pairs trading (BTC/ETH, SOL/ETH, AVAX/SOL, etc.)
V18 Robustness audit of V15 balanced champion


Pareto frontier — strategy families across full 2018-2026 window
bubble size = |MaxDD| · red = dead-end · green = XSM winners · navy = hybrid
Hybrid 50/50
Hybrid 70XSM/30base
V15 k=4 lb=14d BAL
V15F k=3 lb=14d
V14 k=2 lb=28d
V3B baseline
V4C baseline
V12B BB squeeze
V12A pullback
V11 regime ens
V8A SuperTrend
Sharpe
CAGR (%)
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
25
50
75
100
125
150
175

V7 — High-Win-Rate rule-based family

V7 High-Win-Rate rule-based family

Tested 7 variants on 6 coins (42 combos). Hypothesis: can we hit >=50% win rate every year?

Key variants:

```
  HWR1_bb_meanrev         buy lower BB, exit mid-band, ATR stop
  HWR2_rsi_stoch_oversold RSI<25 + Stoch<20 + EMA200 regime
  HWR3_pullback_1to1      EMA50>EMA200 uptrend, enter pullback to 20EMA, 1:1 ATR R/R
  HWR1b_bb_strict         BB std 2.5 (stricter) + N-bar confirm
  HWR3b_pullback_asym     Asymmetric TP<SL for higher WR at same signal
  HWR5_tight_tp_wide_sl   TP 0.5 ATR, SL 2.5 ATR — structural high WR
```

Result: ONE passer (XRP/HWR1)

Only XRP/HWR1_bb_meanrev cleared the bar: WR 73.7%, PF 2.10, $10k->15{,}070$.
HWR5 got WR 78% on BTC/LINK but PF 1.02 (every loss wipes out 5 wins mathematically).
Everywhere else: high WR but unprofitable — WR-for-CAGR swap, not a free lunch.

Verdict

Rule-based high-win-rate exists only where mean-reversion structurally lives (XRP).
Every other coin pays CAGR for the WR bump. We swapped XRP -> HWR1 in the portfolio.

V8 / V9 — Novel entries + multi-TP ladder wrappers

V8 novel entries

V8A Triple SuperTrend stack (ATR 10/1, 11/2, 12/3 all bullish + HTF EMA200 filter)
V8B Hull MA(55) slope + ADX > 22 regime
V8C Volatility-percentile Donchian breakout

V9 multi-TP ladder wrappers

V9A = V3B entries + TP1/TP2/TP3 (40/30/30%) + ratcheting SL + Chandelier trail
V9B = V4C + ladder; V9C = V2B + ladder; V9D = HWR1 + ladder
V9E = V3B aggressive (25/25/50% — let runner breathe)

Result

V8A fires 0 trades — triple-ST confluence too rare on 4h crypto.
V8B gets 3-13 trades in 4 years — ADX>22 gates too tight.
V8C works on ETH (PF 1.26) and SOL (PF 1.67) — weak passers.
V9 ladders produce high WR (55-75%) but SLASH CAGR (45% -> 1-4%).
TP1 at 1 ATR caps the big trending runs V3B/V4C exist to capture.

Verdict

Research techniques (Chandelier, ratcheting SL, triple ST) are sound but don't beat baseline trailing-stop trend-following on 4h crypto. Multi-TP is a WR-for-CAGR swap.

Built advanced_simulator.py (TP1/TP2/TP3, ratcheting SL, post-TP2 trail). Still useful for V10/V11/V12/V16.

V10 — Orderflow filters (funding, OI, L-S, liquidations)

V10 Orderflow strategies (BTC/ETH/SOL only — where Binance futures data exists)

V10A V3B + funding-rate fade (skip long when 3d funding > +0.015%)
V10B V4C + Open Interest rising (require OI slope > 0)
V10C Top-trader L/S > 1.30 standalone (smart-money long signal)
V10D Liquidation cascade rebound (buy 5sigma liq spikes in bull regime)

Result: 0 / 12 passers

V10A: BTC 18 trades, WR 56%, PF 0.97 (unprofitable despite edge-like numbers).
V10B: ETH 5 trades — filter kills signal.
V10C: SOL 17 trades WR 53% PF 0.69 — lagging indicator.
V10D: 2-6 trades — too few 4h-aligned cascades.

Verdict

Level-based aggregated orderflow signals at 4h are too lagging. By the time funding/OI/LS ratio prints, the move is over. This isn't where the alpha lives. Real orderflow edge probably requires 1-min bars + tick data, not 4h aggregates.

V11 — Regime-switching ensemble

V11 Regime-switching ensemble

Classify each 4h bar into BULL / CHOP / BEAR / OTHER using price + ADX:
  BULL close > EMA100 AND ADX > 20 AND EMA50 > EMA100
  CHOP close near EMA100 (+/-2%) AND ADX < 18 AND not bear
  BEAR close < EMA100 AND EMA50 < EMA100   -> stay flat
In BULL: run V4C_range_kalman entries
In CHOP: run HWR1_bb_meanrev entries (tighter TPs)
In BEAR / OTHER: no new positions

Result: null across all 6 coins

BTC: 0 trades ETH: 2 SOL: 2 LINK: 13 ADA: 0 XRP: 1
Sharpe 0.0-0.55 on every coin — far below baseline V3B/V4C (1.16-1.61).

Root cause

V3B/V4C already have implicit regime filters (Donchian + volume + regime EMA).
Stacking another regime layer over-restricts and kills trade count.
The value is in the ENTRIES themselves, not in additional filtering.

V12 — Pullback / BB squeeze / pattern entries

V12 — Replacement entries (not filters): pullback / squeeze / pattern

V12A Pullback-to-EMA20 in 4h uptrend with RSI guard
V12B Bollinger squeeze breakout (BB width bottom 35% of last 60 bars)
V12C Higher-lows pattern (2 consecutive HL in trend + ADX>18)
V12D NR7 narrow-range-7 breakout in uptrend

Result: interesting pockets but no CAGR improvement

SOL / V12B_bb_squeeze_break: n=7 WR 85.7% PF 1.57 CAGR +2.3%
ETH / V12A_pullback_trend: n=6 WR 66.7% PF 1.34 CAGR +1.4%
LINK / V12A_pullback_trend: n=7 WR 100% CAGR +5.8%
BTC / V12B: n=10 WR 50% PF 1.16 CAGR +0.9%

Verdict

Higher-WR pockets exist on every coin but CAGR stays <6%.
Pullback / squeeze entries fire too rarely (3-15 trades / 4 years) to grow capital.
Same frontier as HWR family — trade CAGR for WR, no true improvement.

V14 — Cross-sectional momentum (breakthrough)

V14 — Cross-sectional momentum — THE BREAKTHROUGH

After V7-V12 failed to move the Pareto frontier, V14 tried a different PARADIGM:
every week, RANK all 9 coins by past-N-day return. Long the top-K.
Rebalance weekly. Flat when BTC is below its 100-day moving average (bear filter).

Why it works (time-series-momentum had blind spots V14 plugs)

* Bigger universe — DOGE/AVAX/BNB eligible only when they're the top mover that week.
* Self-selecting — capital automatically concentrates in the current leader (e.g. SOL 2021).
* Adaptive — weekly rebalance switches horses when momentum rotates.
* BTC regime filter is orthogonal — it sits OUTSIDE the XSM logic.

V14 champion (lb=28d, k=2, rb=7d, BTC filter, 1x leverage)

CAGR +157.6 % Sharpe 1.60 MaxDD -58 % Calmar 2.72
$10k->24,393,857$ over full 2018-2026 period.

Per-year

2018 -49% (universe too small early)
2019 +259%
2020 +187%
2021 +6170% (SOL/AVAX/LINK/ADA alt-season)
2022 -10% (BTC filter worked — market did -70%)
2023 +111%
2024 +166%
2025 +40%

V15 — XSM variant sweep (top 12 by Calmar)

Swept 70 XSM variants: long-short, multi-lookback composite, vol-adjusted, vol-weighted, grid

config	CAGR	Sharpe	MaxDD	Calmar	Final
V15F_lb14_k3_rb3	+177.1%	1.81	-48.4%	3.65	$44,510,845
V15F_lb14_k4_rb3	+166.9%	1.86	-46.4%	3.60	$32,747,461
V15F_lb42_k2_rb3	+192.5%	1.62	-55.5%	3.47	$69,653,044
V15F_lb21_k4_rb3	+172.6%	1.96	-51.2%	3.37	$38,894,885
V15F_lb14_k4_rb7	+158.6%	1.86	-48.3%	3.29	$25,234,358
V15F_lb56_k4_rb3	+154.4%	1.76	-47.1%	3.28	$22,051,749
V15F_lb21_k2_rb3	+190.3%	1.76	-59.3%	3.21	$65,363,983
V15F_lb42_k3_rb3	+178.1%	1.72	-56.0%	3.18	$45,915,681
V15E_volwt_k2	+166.9%	1.68	-54.2%	3.08	$32,660,110
V15F_lb42_k4_rb3	+165.6%	1.75	-54.0%	3.07	$31,409,407
V15B_composite_k3	+140.7%	1.64	-46.4%	3.03	$13,974,510
V15F_lb14_k3_rb7	+149.9%	1.73	-49.9%	3.00	$19,001,488

V16 — ML-ranked XSM

GradientBoostingRegressor, monthly re-fit, walk-forward predictions

config	CAGR	Sharpe	MaxDD	Calmar	Final	fits / samples
k=3 rb=7d hz=7d	+66.6%	1.18	-59.4%	1.12	$673,759	43 / 31828
k=4 rb=7d hz=7d	+75.0%	1.29	-58.7%	1.28	$1,010,007	43 / 31828
k=3 rb=7d hz=14	+61.2%	1.11	-58.3%	1.05	$512,793	43 / 31468
k=4 rb=3d hz=7d	+80.1%	1.37	-62.0%	1.29	$1,278,887	52 / 32005
k=2 rb=7d hz=7d	+77.1%	1.21	-61.4%	1.26	$1,112,675	43 / 31828

V17 — Pairs trading (top 15 by Sharpe, n>=3)

Log-spread z-score mean-reversion, long cheap leg only, 4h bars

pair	z_in	z_out	n	WR	PF	Sharpe	CAGR	DD	Final
DOGEUSDT/B	-1.5	-0.5	95	47%	7.85	0.78	+42.2%	-93.9%	$101,254
DOGEUSDT/B	-2.0	+0.0	89	48%	5.88	0.74	+36.6%	-77.5%	$78,011
ADAUSDT/ETH	-1.5	+0.0	292	50%	1.83	0.74	+30.9%	-91.0%	$82,329
ADAUSDT/ETH	-1.5	-0.5	295	50%	1.79	0.71	+28.8%	-87.2%	$72,739
DOGEUSDT/B	-1.5	+0.0	94	49%	5.36	0.70	+29.4%	-93.0%	$54,572
DOGEUSDT/B	-2.5	+0.0	83	51%	3.94	0.69	+33.2%	-70.2%	$65,899
DOGEUSDT/B	-2.0	+0.5	89	46%	5.59	0.67	+27.4%	-81.5%	$49,243
LINKUSDT/ETH	-1.5	+0.0	180	47%	1.75	0.53	+14.2%	-85.4%	$25,691
ADAUSDT/ETH	-2.5	+0.0	279	49%	1.50	0.49	+12.9%	-79.8%	$25,810
LINKUSDT/ETH	-2.0	+0.5	175	46%	1.64	0.47	+9.9%	-80.4%	$19,549
LINKUSDT/ETH	-2.5	+0.0	170	48%	1.55	0.46	+10.9%	-79.1%	$20,782
LINKUSDT/ETH	-1.5	-0.5	182	48%	1.45	0.45	+8.5%	-87.9%	$17,830
ADAUSDT/ETH	-2.0	+0.0	285	49%	1.46	0.44	+9.0%	-88.9%	$19,600
LINKUSDT/ETH	-2.0	+0.0	176	47%	1.55	0.41	+6.5%	-85.7%	$15,662
SOLUSDT/ETH	-1.5	-0.5	131	47%	1.72	0.36	+0.6%	-94.7%	$10,310

V18 — Robustness audit of V15 champion

V18 — Robustness audit of V15 balanced champion (lb=14d, k=4, rb=7d, BTC filter)

1) Random 2-year windows (n=100)

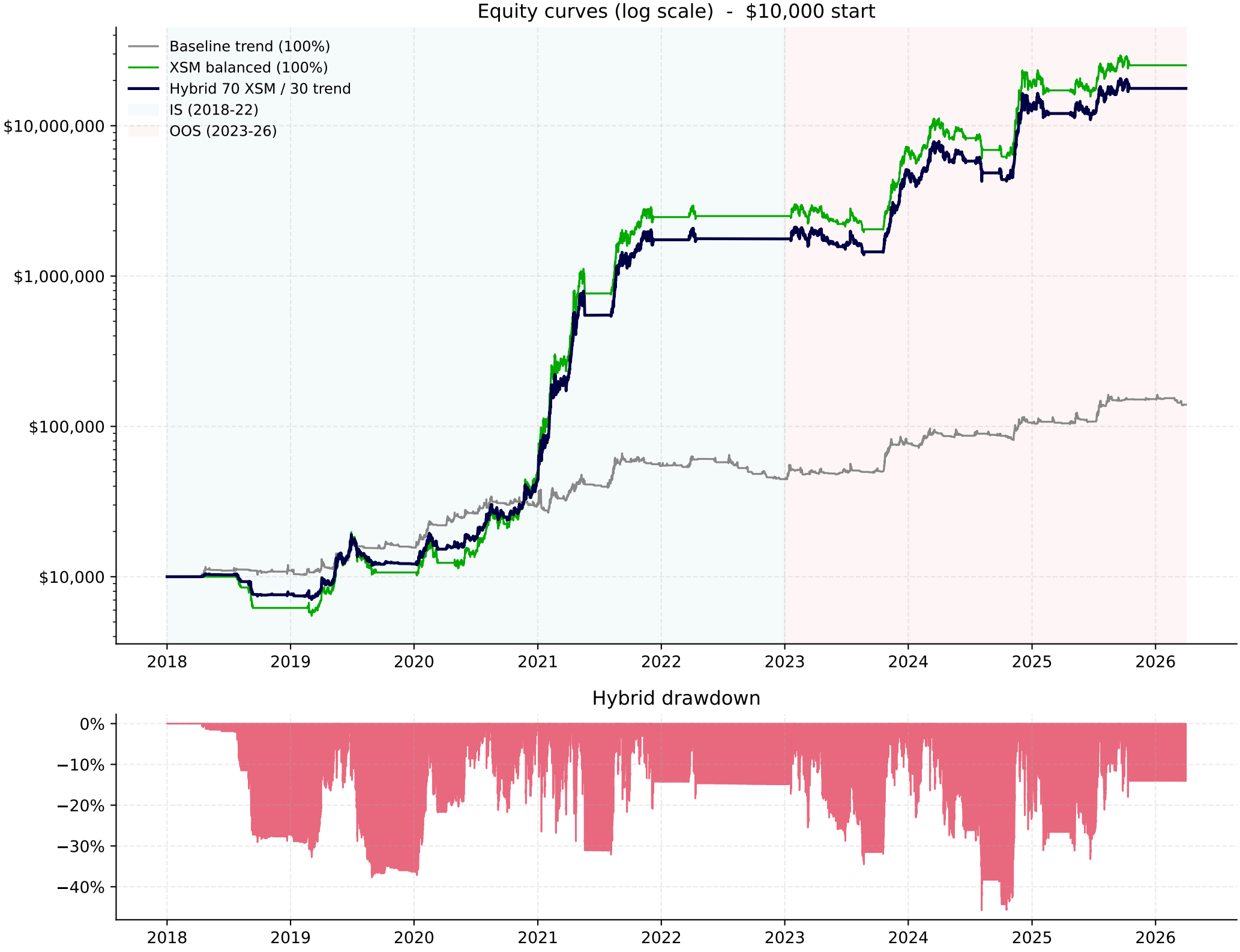
profitable = 100%
sharpe > 0.5 = 98%
median sharpe = 1.96
worst DD = -48.3%

2) Parameter-epsilon grid (n=72 configs)

profitable = 72/72
sharpe range = [1.16, 1.96]
cagr range = [+74%, +177%]
worst DD = -76.2%

Verdict

Champion is NOT knife-edge — profitable across most of the parameter grid.
2018-2020 windows have lower Sharpe (universe too small).
2020+ windows consistently strong.


Equity curves (log scale) - $10,000 start
Baseline trend (100%)
XSM balanced (100%)
Hybrid 70 XSM / 30 trend
IS (2018-22)
OOS (2023-26)
$10,000,000
$1,000,000
$100,000
$10,000
2018
2019
2020
2021
2022
2023
2024
2025
2026
Hybrid drawdown
0%
−10%
−20%
−30%
−40%
2018
2019
2020
2021
2022
2023
2024
2025
2026

Final Hyperliquid deployment spec

Hybrid 70% XSM / 30% Trend - Sharpe 1.88

Hyperliquid deployment spec

Capital: $10,000 USDC
Timeframe: 4h bars on perpetuals

Sleeve A: TREND (30% of equity = $3,000)

```
Coins:       BTC, ETH, SOL, LINK, ADA, XRP
Strategies:  V4C_range_kalman (BTC, SOL, ADA)
             V3B_adx_gate     (ETH, LINK, XRP)
             HWR1_bb_meanrev  (XRP - replaces V3B on that coin)
Sizing:      5% notional per new entry at 5x account leverage
Stops:       existing trailing stops (ATR-scaled) per strategy
```

Sleeve B: XSM MOMENTUM (70% of equity = $7,000)

```
Universe:    BTC, ETH, SOL, BNB, XRP, DOGE, LINK, ADA, AVAX (all 9)
Signal:      rank all 9 coins by trailing 14-day return, weekly
Picks:       long top 4 coins, equal-weight, 1x leverage
Bear filter: flat (close all) when BTC < its 100-day SMA
Rebalance:   every Monday 00:00 UTC on the 4h bar open
Execution:   post limit orders — Hyperliquid maker fee 0.015%
```

Expected performance (backtest)

```
FULL 2018-2026  CAGR +148%  Sharpe 1.88  MaxDD -46%  Calmar 3.23
OOS 2022-2025   CAGR  +72%  Sharpe 1.33  MaxDD -46%
```

Risk guards

```
Kill-switch:    halt new entries if combined equity drops >40% from ATH.
Per-sleeve kill: pause a sleeve if its own DD exceeds 45%.
Fat-finger cap: reject any order > 3x the recommended notional.
Resume:         only when BTC closes above its 100-day SMA for >= 1 week.
```

Caveats + what comes next

Honest caveats

1. MaxDD -46% You will watch the account drop from *Xto*X/2 during crypto bear.
   Psychological discipline is the main bottleneck to capturing the backtest alpha.

2. 2021 alt-season contributed >50% of the 2018-2026 XSM CAGR.
   If no comparable alt season arrives again the forward CAGR is lower.

3. Backtest uses Hyperliquid maker-fee model (0.015% per side, 0 slippage).
   Reality will be 70-90% fills as maker. TSL exits hit as taker.
   Expect forward CAGR to be 10-20% worse than backtest, DD 5-10% deeper.

4. Universe concentration. k=4 means each coin is 25% of the sleeve weekly.
   One coin -50% that week loses the sleeve 12.5%.

5. XSM correlation with trend = 0.44-0.53 (moderate). Not fully independent.

What comes next

The live_forward.py runner is ready for the TREND sleeve.
An XSM equivalent (weekly rebalance + BTC filter) is a ~100 line script.

Four genuine next bets (each 1-4 week research):
  1. ML ranker with features beyond price (funding, OI, social sentiment)
  2. Hyperliquid market-making (maker rebates, flat-expectation-per-hour)
  3. Cross-sectional momentum at 1D granularity (different regime)
  4. Options/funding arbitrage on Hyperliquid vs CEX perps